Remember The Main Idea

Hydrogen fuel-cell vehicles are electric vehicles. They make electricity on board from hydrogen and oxygen instead of charging only from a plug.

Quick Check

Problem
Fossil-fuel dependence

Solution
Hydrogen makes electricity

Best Fit
Buses, trucks, fleets

Cleaner Energy Pathway

E — H_2 — V

Clean energy → **Hydrogen** → **Vehicle**

Low-carbon electricity can produce hydrogen. The climate benefit depends on the full pathway [1], [3].

IEEE Sources

[1] Q. Hassan et al., "Hydrogen Fuel Cell Vehicles: Opportunities and Challenges," Sustainability, vol. 15, no. 15, Art. 11501, 2023, doi: 10.3390/su151511501.
[2] M. Waseem et al., "Fuel cell-based hybrid electric vehicles: An integrated review of current status, key challenges, recommended policies, and future prospects," Green Energy and Intelligent Transportation, vol. 2, no. 6, Art. no. 100121, 2023, doi: 10.1016/j.geits.2023.100121.
[3] C. Gunathilake et al., "A comprehensive review on hydrogen production, storage, and applications," Chemical Society Reviews, vol. 53, no. 22, pp. 10900-10969, 2024, doi: 10.1039/D3CS00731F.
[4] B. Lufkin, "50 grand challenges for the 21st century," BBC Future, 2017. [Online]. Available: https://www.bbc.com/future/article/20170331-50-grand-challenges-for-the-21st-century. [Accessed: Jun. 21, 2026].

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Hydrogen Fuel-Cell Vehicles

A Cleaner Way To Power Transportation

Figure 1. Hydrogen fuel-cell vehicle concept.

H_2 — FC — M — O

Hydrogen → **Fuel cell** → **Motor** → **Motion**

BIG QUESTION

How can vehicles move people and goods while using less fossil fuel?

Electric drive | On-board power | Water at the tailpipe

The Challenge

Many vehicles still burn gasoline or diesel. This creates greenhouse gases and air pollution. Clean transportation means reducing fossil-fuel use while keeping vehicles practical.

Why This Matters

- **C — Cleaner Air**
  Near roads and bus routes.
- **L — Lower Fossil-Fuel Use**
  Less dependence on gasoline and diesel.
- **M — More Engineering Choices**
  Different vehicles need different solutions.

Two Electric Choices

Battery EV
Stores electricity in a battery and charges from the grid.

Fuel-cell EV
Makes electricity on board from stored hydrogen.

What Determines The Best Fit?

- Distance and vehicle duty
- Charging or refuelling time
- Cost and available infrastructure

How The Car Makes Power

Fuel-Cell Flow

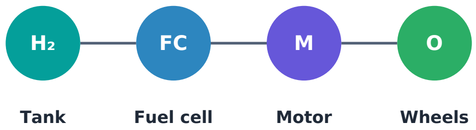


Figure 2. Basic energy flow from hydrogen storage to wheel motion [1].

1. **Store Hydrogen**
   A reinforced tank carries compressed hydrogen.
2. **Bring In Oxygen**
   Air supplies oxygen to the fuel cell.
3. **Make Electricity**
   The electrochemical reaction releases electrons.
4. **Turn The Wheels**
   An electric motor creates motion.

The Chemistry In One Sentence

Hydrogen loses electrons, oxygen combines with hydrogen ions and electrons, and the reaction produces water plus electrical energy.

Source support: [1]

Tailpipe Vs. Full Pathway

At The Vehicle

Water and heat leave the fuel cell.

Before The Vehicle

Production and delivery determine upstream emissions [1], [3].

Why People Are Interested

Clean Tailpipe

Water and heat are the main vehicle outputs.

Quick Refuel

Useful when vehicles cannot wait long.

Longer Routes

Helpful for buses, trucks, and fleets.

Quiet Drive

Electric motors give smooth movement.

Figure 3. Potential benefits for selected transportation uses [1], [2].

Recent reviews highlight fast-refuelling potential and useful range for some transportation jobs [1], [2].

Where Fuel Cells May Fit Best

- B **Buses**
  City or intercity routes.
- H **Heavy Trucks**
  Long routes and high daily use.
- D **Depot Fleets**
  Predictable routes and central refuelling.
- T **Time-Sensitive Operations**
  Downtime affects productivity.

The best technology depends on the job. A short daily commute and a long-haul truck do not have the same energy needs.

Why Fleets Are Practical Test Cases

Known routes make fuel demand easier to plan.

A central depot can serve many vehicles.

High daily use makes refuelling time more important.

What Still Needs Work

Main Barriers

- S **Stations**
  More refuelling locations are needed.
- C **Clean Hydrogen**
  The fuel must be produced with low-carbon energy.
- C **Cost**
  Fuel-cell systems remain expensive.
- S **Storage**
  Tanks require careful safety design.

Takeaway

Hydrogen fuel cells are not a perfect answer, but they are a serious engineering option for cleaner transportation.

Questions Engineers Must Answer

- H **Hydrogen Source**
  Where will the fuel come from?
- E **Energy Pathway**
  How much energy is used before the vehicle?
- S **Station Network**
  Can enough refuelling sites be built?
- B **Best Application**
  Which vehicle types gain the most?

These questions connect vehicle design with energy production, infrastructure, cost, and public safety [1]-[3].

Engineering Claim

Hydrogen fuel-cell vehicles should be treated as a targeted clean-transport option, not a universal replacement for all vehicles [1]-[3].

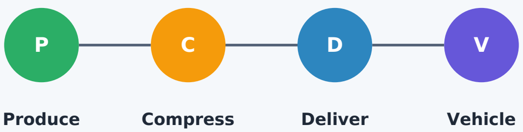
Best-Fit Use Cases

Central-Depot Fleets
High utilization and controlled refuelling.

Heavy Or Long-Route Service
Fast refuelling can reduce downtime.

Low-Carbon Supply Regions
Fuel production determines lifecycle impact.

Evaluate The Full Pathway



IEEE Sources

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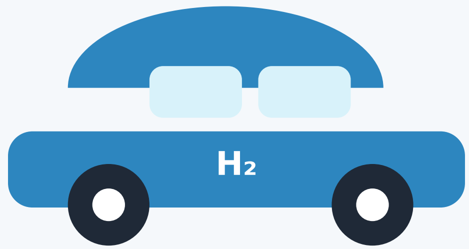
Hydrogen Fuel-Cell Vehicles

PEM Fuel Cells For Lower-Carbon Transportation



SCOPE

Reduce fossil-fuel dependence through fuel-cell electric vehicles and supporting hydrogen infrastructure.

DESIGN OBJECTIVE

Balance range, refuelling time, durability, safety, lifecycle emissions, and total cost.

Vehicle performance cannot be separated from the fuel pathway.

Why This Technology?

Fuel-cell vehicles convert chemical energy into electricity on board. They drive like EVs, but performance also depends on hydrogen production, compression, storage, distribution, and safety.

Key System Boundary

Vehicle
Tanks, stack, battery buffer, motor, controls.

Fuel Pathway
Source, compression, transport, refuelling.

Impact
Well-to-wheel emissions, cost, safety, durability.

Well-To-Wheel Pathway

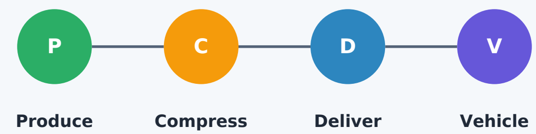


Figure 5. Well-to-wheel hydrogen pathway. Each stage changes energy use, cost, and emissions [1], [3].

Evidence Summary

Recent reviews identify infrastructure, cost, storage, and hydrogen pathway emissions as major adoption barriers [1]-[3].

PEMFC Operation

PEM Fuel-Cell Reaction

ANODE	$H_2 \rightarrow 2H^+ + 2e^-$
CATHODE	$\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$
OVERALL	$2H_2 + O_2 \rightarrow 2H_2O$ + power + heat

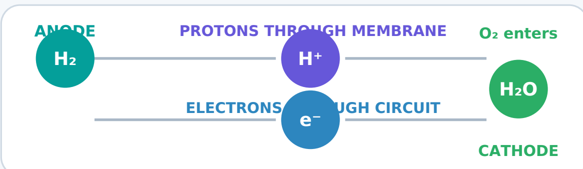


Figure 6. Protons cross the membrane while electrons travel through the external circuit [1].

At the anode, hydrogen oxidation releases protons and electrons. Oxygen reduction at the cathode produces water [1].

Primary Design Variables

Water + Heat

Maintain membrane hydration and temperature.

Catalyst

Limit loading and degradation.

Membrane

Protect durability and gas separation.

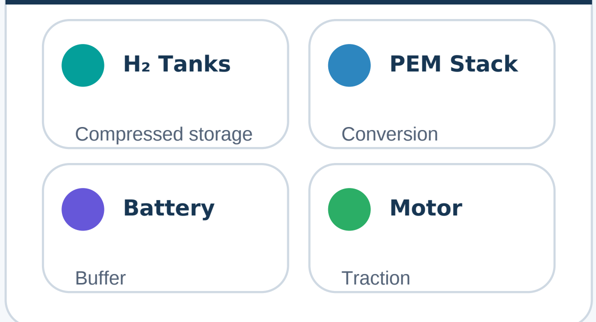
Transient Load

Manage rapid demand and cold start.

These variables influence efficiency, lifetime, packaging, control strategy, and cost.

System Architecture

Figure 7. Vehicle Architecture [2]



A practical vehicle combines a fuel-cell stack with a small battery or supercapacitor. The buffer supports acceleration, regenerative braking, and load transients while the stack supplies steadier power [2].

Energy-Management Priorities

1. **Steady Stack Operation**
   Stay near efficient operating points.
2. **Battery Support**
   Handle peaks and regenerative braking.
3. **Life Protection**
   Limit damage during rapid load changes.
4. **Balance Of Plant**
   Control pressure, temperature, and water.

Control Objective

Meet driver demand while limiting hydrogen use and component degradation.

Barriers And Outlook

Adoption Barriers

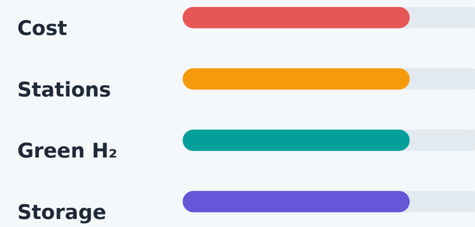


Figure 8. Common adoption barriers identified in recent reviews [1]-[3].

Compressed hydrogen enables onboard storage, but it adds tank mass, safety requirements, and infrastructure cost [1], [3].

Engineering Outlook

A practical near-term strategy is to deploy vehicles where one operator controls both the fleet and refuelling network. Buses, trucks, and depot fleets are strong early candidates [1], [2].

Decision Factors

Duty Cycle

Annual use and route demands.

Hydrogen Supply

Carbon intensity and availability.

Station Case

Utilization and capital cost.

Operations

Durability, safety, and maintenance.

A sound decision compares the complete hydrogen pathway against other low-carbon options for the same transportation job.